

Enumeration explained.

The phase where open ports stop being numbers and start becoming real information — turning a scan into a usable profile of the target.

- ▶ What enumeration actually is, and where it sits in the recon lifecycle
- ▶ 9 services worth targeting, and what each one leaks
- ▶ 5 tools worth running before you even touch Nmap
- ▶ The Nmap commands and flags that matter most

What is Enumeration?

Enumeration is the phase that comes right after **port scanning** in the ethical hacking lifecycle. Where scanning tells you *which ports are open*, enumeration answers the next question: *what exactly is running there, and what can I learn from it?*

An attacker or pentester makes deliberate, active connections to a target's live services to pull out detailed information — the goal being to build a profile of the target that eventually supports gaining access.

The recon lifecycle

- ▶ **Footprinting / OSINT** — passive info gathering
- ▶ **Scanning** — find live hosts and open ports
- ▶ **Enumeration** — actively query those open ports for detail
- ▶ **Exploitation** — use what was found to gain access

What enumeration can reveal

- ▶ Network shares and mounted drives
- ▶ Usernames and (sometimes) passwords
- ▶ Application / service version numbers
- ▶ Users, groups, and machine names
- ▶ Service configuration and security settings
- ▶ Other reachable network resources

Commonly enumerated services

#	Service	Port	What it typically leaks
1	SMTP	25	Valid usernames on the mail server
2	DNS	53	Hostnames, subdomains, IPs, sometimes usernames
3	Microsoft RPC	135	Endpoint mapper info, exposed RPC services
4	NetBIOS Name Service	137	Machine names, workgroup/domain info
5	NetBIOS Session (SMB over NetBIOS)	139	Shares, sessions
6	SMB over TCP	445	Shares, users, OS details, password policy, null sessions
7	NTP	123	Connected hosts, internal IP addresses
8	SNMP	161	Hosts, routes, ARP/routing tables, network resources
9	LDAP	389	Usernames, addresses, directory (e.g. Active Directory) details

Service notes

- ▶ **SMTP** — used to send mail; probing it (e.g. VRFY/EXPN) can confirm valid users.
- ▶ **DNS** — zone transfers or record enumeration expose internal hostnames and structure.
- ▶ **NetBIOS** — lets machines on a LAN share files/printers; reveals computer lists, shares, policies.
- ▶ **SMB** — the TCP-native evolution of NetBIOS sharing (dedicated port 445 since Windows 2000); reveals shares, users, OS, groups, and null-session weaknesses.
- ▶ **NTP** — keeps clocks in sync; querying it can map connected hosts and internal addressing.
- ▶ **SNMP** — weak community strings expose hosts, routes, and ARP tables.
- ▶ **LDAP** — the protocol behind directory services like Active Directory; can surface valid usernames.

Enumeration tools

theHarvester

OSINT tool for pulling emails, subdomains, hosts, and names tied to a target domain from public sources.

dmitry

Deepmagic Information Gathering Tool — passive info gathering: whois, subdomains, emails, open ports.

whatweb

Fingerprints a website's tech stack (like Wappalyzer, but from the CLI) — CMS, frameworks, servers, plugins.

```
whatweb nixsecura.com
```

wafw00f

Detects whether a site is protected by a Web Application Firewall, and tries to identify which one.

```
wafw00f tesla.com # shows the WAF/web applications running on it
```

spiderfoot

OSINT automation framework with a web GUI. Runs locally and pulls together data from dozens of sources.

```
spiderfoot -l 127.0.0.1:9000 # -l = local IP:port to serve the GUI on  
# then target a domain from the GUI, e.g. hackthissite.org
```

Nmap for enumeration

Once a target is confirmed live, Nmap is the go-to tool to enumerate open ports, service versions, and OS details.

<code>nmap <IP></code>	Basic scan — top 1000 TCP ports
<code>nmap -sV <IP></code>	Service/version detection on open ports
<code>nmap -sC <IP></code>	Runs Nmap's default safe NSE scripts
<code>nmap -A <IP></code>	Aggressive scan — OS + version + scripts + traceroute
<code>nmap -O <IP></code>	OS fingerprinting
<code>nmap -sV <IP> -oN result.txt</code>	Saves scan output to a normal .txt file
<code>nmap -sV <IP> -oA filename</code>	Saves output in all 3 formats: .nmap, .gnmap, .xml

Handy combos:

`nmap -sC -sV -T4 target` — the most common quick recon command

`nmap -p- -sV target` — service-version scan across all 65535 ports

`nmap -p 445 --script smb-enum-shares,smb-enum-users target` — targeted SMB enumeration

`nmap -p 80,443 --script http-enum,http-title,http-headers target` — quick web service enumeration

Scope & legality: every command above should only be run against systems you own or are explicitly authorised to test (lab VMs, CTF platforms like HackThisSite/TryHackMe/HackTheBox, or written client authorisation). Unauthorised scanning is illegal in most jurisdictions.